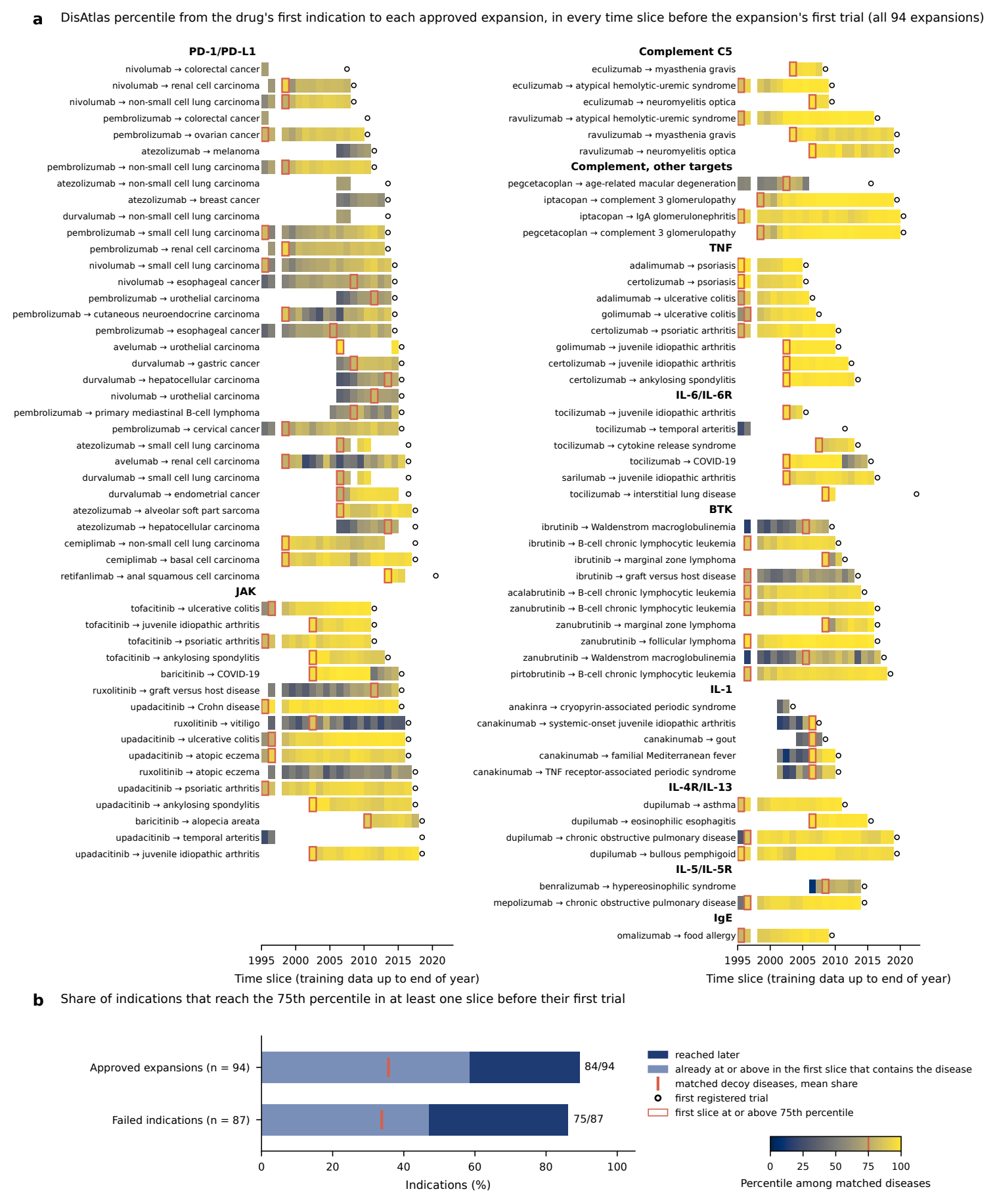

a
DisAtlas percentile from the drug's first indication to each approved expansion, in every time slice before the expansion's first trial (all 94 expansions)
PD-1/PD-L1
nivolumab → colorectal cancer
nivolumab → renal cell carcinoma
nivolumab → non-small cell lung carcinoma
pembrolizumab → colorectal cancer
pembrolizumab → ovarian cancer
atezolizumab → melanoma
pembrolizumab → non-small cell lung carcinoma
atezolizumab → non-small cell lung carcinoma
atezolizumab → breast cancer
durvalumab → non-small cell lung carcinoma
pembrolizumab → small cell lung carcinoma
pembrolizumab → renal cell carcinoma
nivolumab → small cell lung carcinoma
nivolumab → esophageal cancer
pembrolizumab → urothelial carcinoma
pembrolizumab → cutaneous neuroendocrine carcinoma
pembrolizumab → esophageal cancer
avelumab → urothelial carcinoma
durvalumab → gastric cancer
durvalumab → hepatocellular carcinoma
nivolumab → urothelial carcinoma
pembrolizumab → primary mediastinal B-cell lymphoma
pembrolizumab → cervical cancer
atezolizumab → small cell lung carcinoma
avelumab → renal cell carcinoma
durvalumab → small cell lung carcinoma
durvalumab → endometrial cancer
atezolizumab → alveolar soft part sarcoma
atezolizumab → hepatocellular carcinoma
cemiplimab → non-small cell lung carcinoma
cemiplimab → basal cell carcinoma
retifanlimab → anal squamous cell carcinoma
JAK
tofacitinib → ulcerative colitis
tofacitinib → juvenile idiopathic arthritis
tofacitinib → psoriatic arthritis
tofacitinib → ankylosing spondylitis
baricitinib → COVID-19
ruxolitinib → graft versus host disease
upadacitinib → Crohn disease
ruxolitinib → vitiligo
upadacitinib → ulcerative colitis
upadacitinib → atopic eczema
ruxolitinib → atopic eczema
upadacitinib → psoriatic arthritis
upadacitinib → ankylosing spondylitis
baricitinib → alopecia areata
upadacitinib → temporal arteritis
upadacitinib → juvenile idiopathic arthritis
1995 2000 2005 2010 2015 2020
Time slice (training data up to end of year)
Complement C5
eculizumab → myasthenia gravis
eculizumab → atypical hemolytic-uremic syndrome
eculizumab → neuromyelitis optica
ravulizumab → atypical hemolytic-uremic syndrome
ravulizumab → myasthenia gravis
ravulizumab → neuromyelitis optica
Complement, other targets
pegcetacoplan → age-related macular degeneration
iptacopan → complement 3 glomerulopathy
iptacopan → IgA glomerulonephritis
pegcetacoplan → complement 3 glomerulopathy
TNF
adalimumab → psoriasis
certolizumab → psoriasis
adalimumab → ulcerative colitis
golimumab → ulcerative colitis
certolizumab → psoriatic arthritis
golimumab → juvenile idiopathic arthritis
certolizumab → juvenile idiopathic arthritis
certolizumab → ankylosing spondylitis
IL-6/IL-6R
tocilizumab → juvenile idiopathic arthritis
tocilizumab → temporal arteritis
tocilizumab → cytokine release syndrome
tocilizumab → COVID-19
sarilumab → juvenile idiopathic arthritis
tocilizumab → interstitial lung disease
BTK
ibrutinib → Waldenstrom macroglobulinemia
ibrutinib → B-cell chronic lymphocytic leukemia
ibrutinib → marginal zone lymphoma
ibrutinib → graft versus host disease
acalabrutinib → B-cell chronic lymphocytic leukemia
zanubrutinib → B-cell chronic lymphocytic leukemia
zanubrutinib → marginal zone lymphoma
zanubrutinib → follicular lymphoma
zanubrutinib → Waldenstrom macroglobulinemia
pirtobrutinib → B-cell chronic lymphocytic leukemia
IL-1
anakinra → cryopyrin-associated periodic syndrome
canakinumab → systemic-onset juvenile idiopathic arthritis
canakinumab → gout
canakinumab → familial Mediterranean fever
canakinumab → TNF receptor-associated periodic syndrome
IL-4R/IL-13
dupilumab → asthma
dupilumab → eosinophilic esophagitis
dupilumab → chronic obstructive pulmonary disease
dupilumab → bullous pemphigoid
IL-5/IL-5R
benralizumab → hypereosinophilic syndrome
mepolizumab → chronic obstructive pulmonary disease
IgE
omalizumab → food allergy
1995 2000 2005 2010 2015 2020
Time slice (training data up to end of year)
b
Share of indications that reach the 75th percentile in at least one slice before their first trial
Approved expansions (n = 94)
84/94
Failed indications (n = 87)
75/87
0 20 40 60 80 100
Indications (%)
reached later
already at or above in the first slice that contains the disease
matched decoy diseases, mean share
first registered trial
first slice at or above 75th percentile
0 25 50 75 100
Percentile among matched diseases